Module 1
Introduction to Database and Data Modeling

Co1:Understand concepts of DBMS and Design ER/EER diagram for real world application.

Class: SE CSE(AI & ML)

Sem: IV

What is Data?

A collection of raw unprocessed figures, facts and symbol.

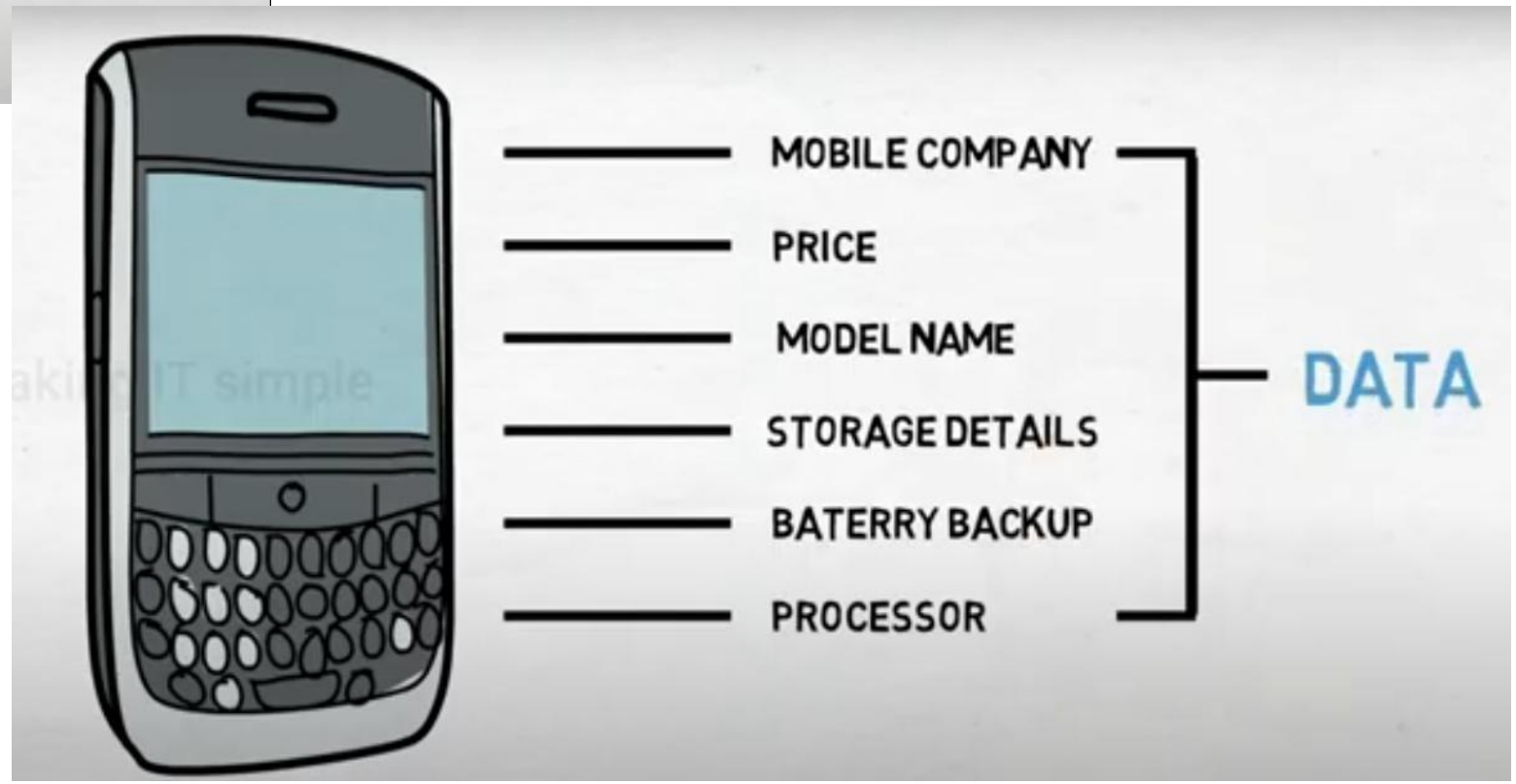
e.g Priya , 22, Ratnagiri, etc

What is Information?

Information is processed, organized, or structured data that is meaningful and useful.

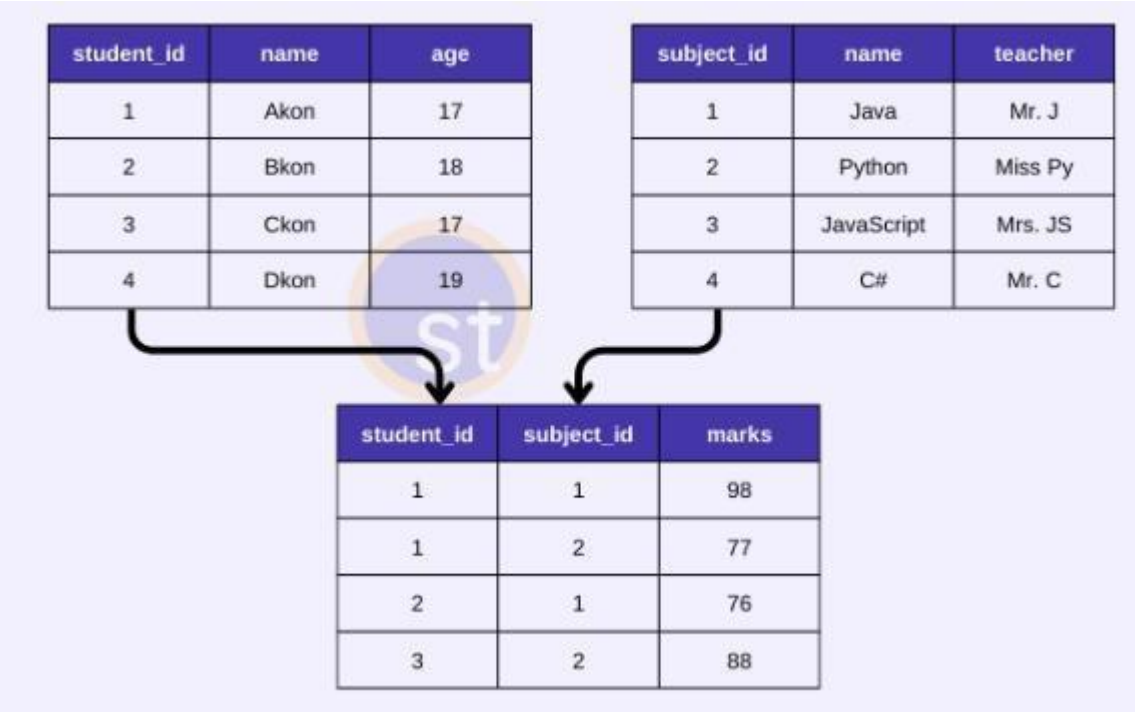
e.g Priya is 22yrs old and lives in Ratnagiri.

NAME
AGE
HEIGHT
WEIGHT
MOB NO


MOBILE COMPANY
PRICE
MODEL NAME
STORAGE DETAILS
BATERRY BACKUP
PROCESSOR
DATA

What is Database?

- A database is an organized collection of structured information, or data, typically stored electronically in a computer system.
- **A Database is a structured collection of related data that can be easily accessed, managed, and updated**.
- A database is usually controlled by a database management system (DBMS).



student_id	name	age
1	Akon	17
2	Bkon	18
3	Ckon	17
4	Dkon	19

subject_id	name	teacher
1	Java	Mr. J
2	Python	Miss Py
3	JavaScript	Mrs. JS
4	C#	Mr. C

student_id	subject_id	marks
1	1	98
1	2	77
2	1	76
3	2	88

College Database

Department

Dept_ID	Dept_Name
1	IT
2	CSE
3	MECH
4	EXTC

Student

Student_ID	Stud_Name	Class	Dept_ID
1	Smith	SE	1
2	Alex	TE	2
3	John	SE	3
4	David	TE	4
5	Priya	BE	1
6	Alice	TE	2
7	Mark	BE	1

What is Database Mangement System?

- A DBMS is a software that allows creation, definition and manipulation of database, allowing users to store, process and analyse data easily.
- It acts as a bridge between users and the data.
- Some examples of popular DBMS used these days:
    - MySql
    - Oracle
    - SQLite
    - MongoDB
    - IBM DB2
    - PostgreSQL

Name
Rajesh Kumar
Mobile No.
9876543210
SSC Marks
450
HSC Marks
410
Address
123, MG Road
SUBMIT
Name
Mobile No.
SSC Marks
Address
Rajesh
9876543210
450
123, MG Road

Fee_Details

RegNO	Student Name	Branch	Mobile No	Class	Fees	Fees Paid	Fee Balance
1	Alice	IT	899999999	SE	50000	20000	30000
2	David	IT	950000000	SE	84000	80000	4000
3	Mark	Mech	999999999	SE	50000	50000	0

Academic_Details

RegNO	Student Name	Branch	Mobile No	Class	Sub1	Sub2	Sub3
1	Alice	IT	899999999	SE	45	67	89
2	David	IT	950000000	SE	54	78	46

Student_Details

RegNO	Student Name	Branch	Mobile No	Class
1	Alice	IT	899999999	SE
2	David	IT	950000000	SE
3	Mark	Mech	999999999	SE

Fee_Details

RegNO	Fees	Fees Paid	Fee Balance
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2	84000	80000	4000
3	50000	50000	0

Academic_Details

RegNO	Class	Sub1	Sub2	Sub3
1	SE	45	67	89
2	SE	54	78	46

What are the Characterstics of Databases?

1. Data Persistence

- Data is stored permanently and remains available until explicitly deleted.
- Even after power loss, data doesn’t vanish.

2. Data Integrity

- Ensures the accuracy and consistency of data.
- Example: If a student's roll number must be unique, the database enforces that rule.

3. Data Security

- Access control ensures only authorized users can view or modify data.
- Prevents unauthorized access or accidental changes.

4. Minimal Data Redundancy

- Avoids storing the same data in multiple places.
- Saves space and reduces errors caused by duplicate values.

5. Data Consistency

- Ensures that all copies or references of data are consistent across the system.
- Example: If a student's name is updated, it reflects everywhere in the database.

6. Data Independence

- Applications remain unaffected if the structure of the data changes at the storage level.
- Logical (what users see) and physical (how data is stored) levels are separate.

7. Efficient Query Processing

- Supports fast retrieval and modification of data using query languages like SQL.

8. Multiple User Access

- Multiple users can access and modify data at the same time without interference.
- Handled through concurrency control.

9. Backup and Recovery

- Automatic or manual backups prevent data loss.
- Recovery tools help restore data after a system crash.

What is File System?

- The file system is basically a way of arranging the files in a storage medium like a hard disk.
- The file system organizes the files and helps in the retrieval of files when they are required.
- File systems consist of different files which are grouped into directories. The directories further contain other folders and files.
- The file system performs basic operations like management, file naming, giving access rules, etc.

What are Limitations of File System ?

1. **Separated and Isolated Data:**
   - To take decisions, the user might need to access data from two separate files.
   - For example, if a user wants academic information and the fee of a student, then he needs two different files, i.e. general office and account office. We cannot access all the data from a single file, as it separately stores data for each file.
2. **Duplication of Data / Data Redundancy:**
   - Often, the same information is stored in multiple files and it costs more time and money.
   - As in the university database, name, class, roll no is the same information of a student that is stored in different files. So, in a file-based system, data duplication is there.
3. **Data Inconsistency :**
   - Data inconsistency mean that different files may contain different information of a particular object or person. Actually redundancy leads to inconsistency.When the same data is stored in multiple locations, the inconsistency may occur.

4. **Poor Data Security:**
   - In the file-based system the security of data is low because data is maintained in files and is easily accessible.
   - And there are no security mechanisms to protect the files.

5. **Transaction Problem:**
   - Cannot guarantee ACID properties (Atomicity, Consistency, Isolation, Durability).

6. **No Backup and Recovery:**
   - File system doesn’t provide the facility of recovery, in case of data loss.

7. **Concurrency Problem:**
   - When multiple users access the same piece of data at the same interval of time, then it is called the concurrency of the system.
   - And if they want to update a file simultaneously, then a concurrency problem occurs in a file-based system.

Compare File System vs DBMS

File System	DBMS
The file system is a way of arranging the files in a storage medium within a computer.	DBMS is software for managing the database.
Redundant data can be present in a file system.	In DBMS there is no redundant data.
It doesn't provide Inbuilt mechanism for backup and recovery of data if it is lost.	It provides in house tools for backup and recovery of data even if it is lost.
File systems provide less security in comparison to DBMS.	DBMS has more security mechanisms as compared to file systems.
There is no data independence in file system.	Data independence exists in DBMS.
Only one user can access data at a time.	Multiple users can access data at a time
It doesn’t have concurrency control.	DBMS has concurrency control.
There is less data consistency in the file system.	There is more data consistency because of the process of normalization .

What is Data Abstraction?

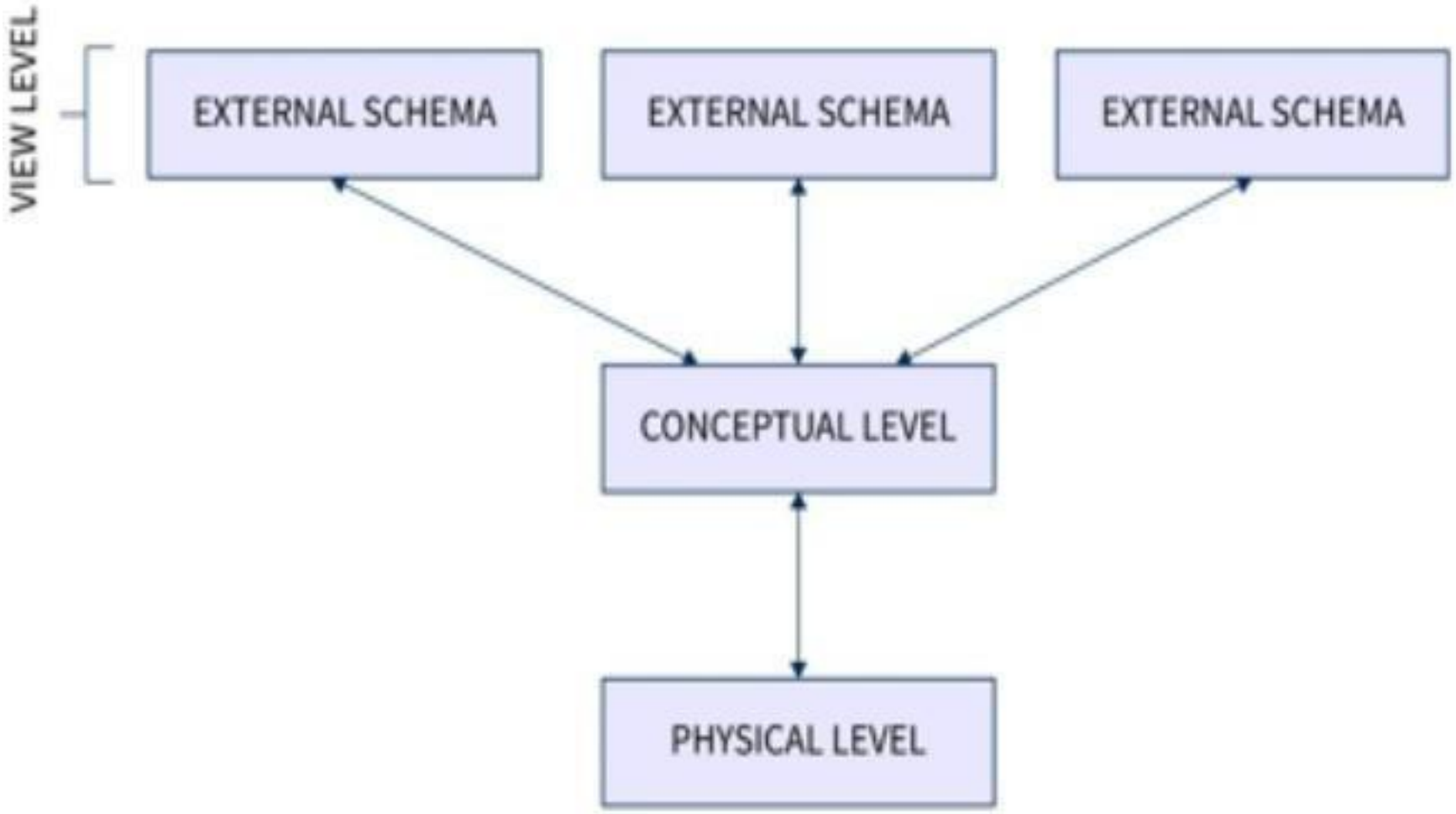
- Data Abstraction in DBMS is the **process of hiding the complex details of how data is stored and maintained**, while showing only the necessary parts to users.

- It helps **simplify interaction with the database and supports data independence**—meaning changes in storage or structure don't affect how users access the data.

- With the help of data abstraction, **developers hide irrelevant data from the user and provide them the relevant data**.

- By doing this, users can access the data without any hassle, and the system will also work efficiently.

- **In DBMS, data abstraction is performed in layers : Physical Level, Conceptual Level and View Level**


VIEW LEVEL
EXTERNAL SCHEMA
EXTERNAL SCHEMA
EXTERNAL SCHEMA
CONCEPTUAL LEVEL
PHYSICAL LEVEL

1. **Physical Level:**
   - It is the layer that **defines how data is actually stored in the database**.
   - It **defines methods to access the data** in the database.
   - It **defines complex data structures in detail**, so it is very complex to understand, which is why it is kept hidden from the end user.
   - Data Administrators (DBA) decide how to arrange data and where to store data. The Data Administrator (DBA) is the person whose role is to manage the data in the database at the physical or internal level.

2. **Conceptual Level:**
   - It **explains what data is going to be stored in the database and what the relationship is between them**.
   - It **describes the structure of the entire data** in the form of tables.
   - The logical level or conceptual level is **less complex** than the physical level.
   - With the help of the logical level, Data Administrators (DBA) abstract data from raw data present at the physical level.

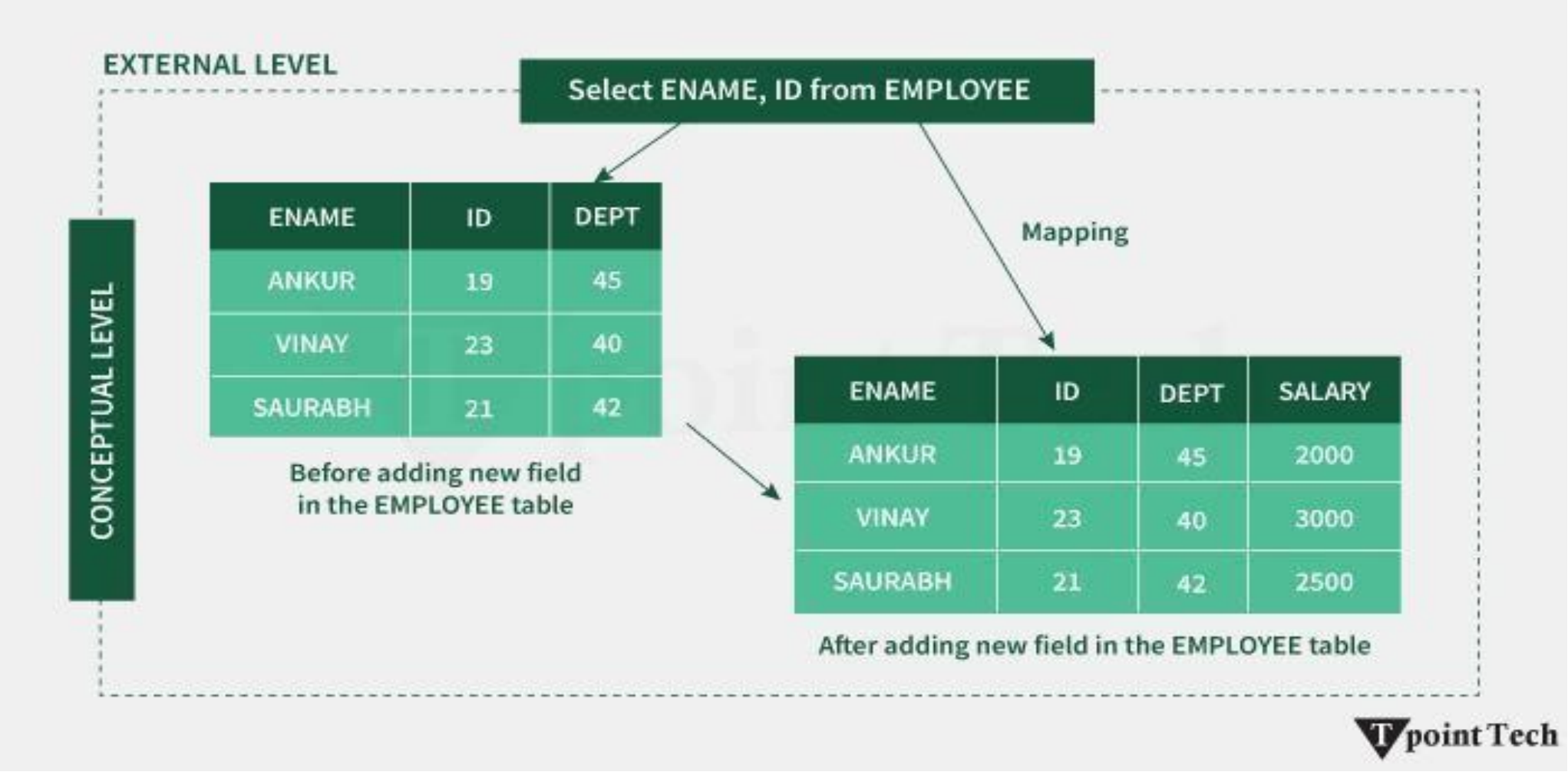
3. View Level:
   - This level is for the **end-user interaction**; at this level, **end users can access the data based on their queries.**
   - End-users does not get the entire database, but depending on the queries made from the front-end the user gets to see the data.
   - It may be a single data from the entire database or a collection of data in tabular format.
   - **Multiple views of the same data are available to the user,** the representation can be a table, a graph, or a pie chart.
   - View Level is **the highest level of data abstraction** in DBMS.

What is Data Independence?

- Data independence refers characteristic of being able to modify the schema at one level of the database system without altering the schema at the next higher level.
- There are two types of data independence:
    1. Logical Data Independence
    2. Physical Data Independence

1. Logical Data Independence

- **Changing the logical schema (conceptual level) without changing the external schema (view level)** is called logical data independence.
- It is **used to keep the external schema separate from the logical schema**.
- If we make any changes at the conceptual level of data, it does not affect the view level.
- This happens at the user interface level.
- For example, it is possible to add or delete new entities, attributes to the conceptual schema without making any changes to the external schema.


EXTERNAL LEVEL
Select ENAME, ID from EMPLOYEE
CONCEPTUAL LEVEL
Mapping
ENAME ID DEPT
ANKUR 19 45
VINAY 23 40
SAURABH 21 42
Before adding new field in the EMPLOYEE table
ENAME ID DEPT SALARY
ANKUR 19 45 2000
VINAY 23 40 3000
SAURABH 21 42 2500
After adding new field in the EMPLOYEE table
Tpoint Tech

2. Physical Data Independence

- Making c**hanges to the physical schema without changing the logical schema** is called physical data independence.
- If we change the storage size of the database system server, it will not affect the conceptual structure of the database.
- It is **used to keep the conceptual level separate from the internal level**.
- This happens at the logical interface level.
- Example – Changing the location of the database from C drive to D drive.

What is the role of Database Abministrator (DBA)?

- A Database Administrator is a specialized IT professional responsible for managing databases, which serve as the backbone of data-oriented applications.
- They ensure data is stored securely, accessible to the right users, and performing optimally.
- A DBA can be a system administrator who was given the added responsibility of maintaining a SQL Server.
- A DBA can be a programmer who, by default or by volunteering, took over the responsibility of maintaining a SQL Server during project development

Roles and Responsiblities of DBA

1. Database Installation and Configuration

- Set up database software (like Oracle, MySQL, SQL Server).
- Configure settings for optimal performance and security.

2. Database Design and Implementation

- Help design the structure of databases (tables, relationships, constraints).
- Ensure the design supports business needs and scalability.

3. Performance Monitoring and Tuning

- Monitor database speed and responsiveness.
- Optimize queries and indexing to improve performance.

4. Backup and Recovery

- Set up regular backups to prevent data loss.
- Restore data in case of system failure or corruption.

5. Security Management

- Control user access and permissions.
- Assigning permissions to the various database objects such as tables, views, and stored procedures
- Protect data from unauthorized access or breaches.

6. Data Integrity and Consistency

- Enforce rules to ensure data accuracy and reliability.
- Prevent duplication and maintain consistency across systems.

7. Troubleshooting and Support

- Resolve database errors and issues.
- Provide support to developers and users.

8. Providing 24-Hour Access:

- the databases must always be protected and online.
- Be prepared to perform some maintenance and upgrades after hours.
- If the database server should go down, be ready to get the server up and running.

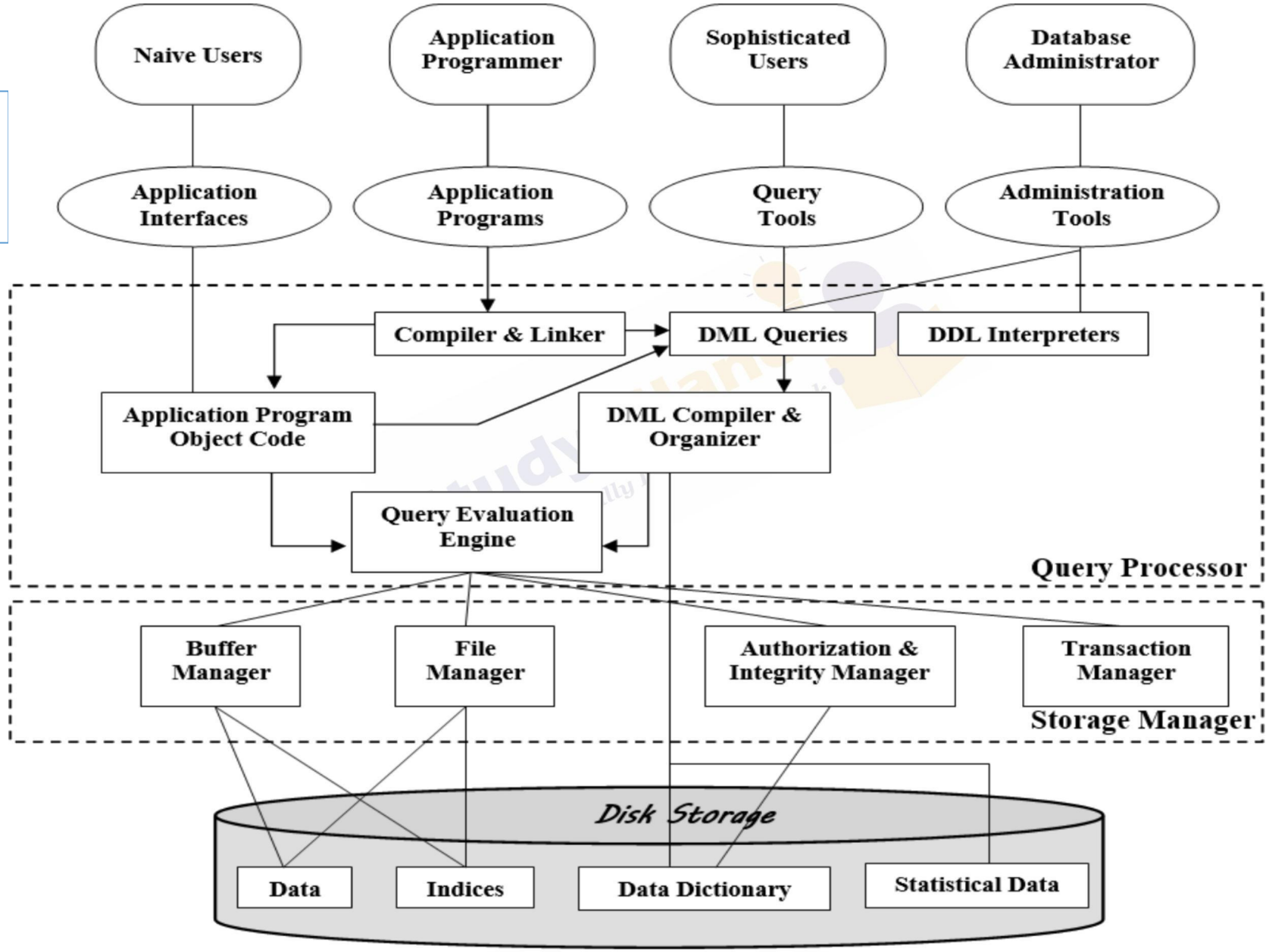
9. Upgrades and Patching

- Apply updates to database software to fix bugs or add features.
- Ensure minimal downtime during upgrades.

10. Documentation and Reporting

- Maintain records of configurations, procedures, and changes.
- Generate reports for audits or performance reviews.

Overall DBMS Architecture



I. SYSTEM USERS

There are four different types of database-system users,

1. Application programmers
2. Sophisticated users and Specialized users:-
3. Database Administrator
4. Naive users

1. Application programmers

- Computer professionals and write application programs.
- They interact with the system through DML calls.
- DML calls are embedded in a program written in a host language (for example PHP, python, c).
- These programs are commonly referred to as application programs.

2. Sophisticated users and Specialized users:

- Interact with the system without writing programs.
- They request in a database query language as query form.
- Each query is submitted to a query processor.
- Query processor is to break down DML statements into instructions that the storage manager understands.
- They work as an analyst.
- Specialized users are sophisticated users who write specialized database applications like computer-aided design Systems, knowledgebase, and expert systems, etc.
- The user uses complex data types (for example, graphics data and audio data).

3. Database Administrator (DBA)

- DBA has central control over the system.
- Responsible for following functions:

(i) Schema Design and Maintenance,
(ii) Physical Schema and Organization Modification,
(iii) Authorization and Security,
(iv) Integrity Constraint Specification,
(v) Recovery from Failure,
(vi) Database Upgradation

4. Naive users

- Naive users are unsophisticated users.
- Interact with the system by the permanent application programs that have been written previously.
- For example, the clerk at the ticket booking window, he uses an application program to do his job of making reservations for a passenger.

II. QUERY PROCESSOR:

- The interactive query processor helps the database system to simplify and facilitate access to data.
- The following are various functionalities and components of query processor

1. **DDL interpreter:** This is basically a translator which interprets the DDL statements in data dictionaries.
2. **DML compiler:** It translates DML statements query language into an evaluation plan. This plan consists of the instructions which query evaluation engine understands.
3. **Query evaluation engine**: It executes the low-level instructions generated by the DML compiler.

- When a user issues a query, the parsed query is presented to a query optimizer, which uses information about how the data is stored to produce an efficient execution plan for evaluating the query.
- An execution plan is a blueprint for evaluating a query. It is evaluated by query evaluation engine.

III. STORAGE MANAGER:

- Storage manager is the component of database system that provides interface between the low level data stored in the database and the application programs and queries submitted to the system.
- The storage manager is responsible for storing, retrieving, and updating data in the database. The storage manager components include

1. **Authorization and integrity manager:** Validates the users who want to access the data and tests for integrity constraints.
2. **Transaction manager:** Ensures that the database remains in consistent despite of system failures and concurrent transaction execution proceeds without conflicting.

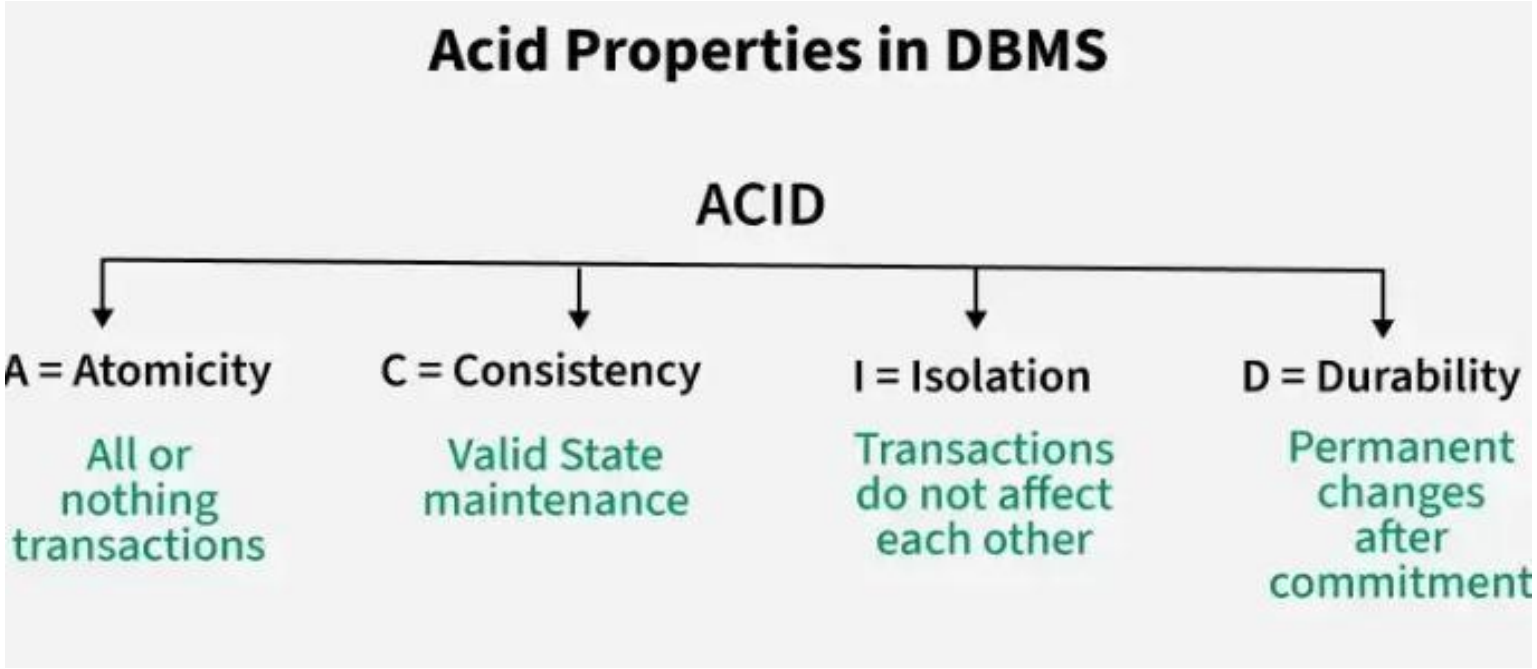
3. **File manager:** Manages allocation of space on disk storage and representation of the information on disk.
4. **Buffer manager:** Manages the fetching of data from disk storage into main memory. The buffer manager also decides what data to cache in main memory. Buffer manager is a crucial part of database system.

Storage manager implements several data structures such as

1. Data files: Used for storing database itself.
2. Data dictionary: Used for storing metadata, particularly schema of database.
3. Indices: Indices are used to provide fast access to data items present in the database

Explain ACID property in DBMS?

- **For maintaining the integrity of data in the database** the certain properties are followed by all the transactions that take place in the database.
- These properties are popularly known as ACID properties where
  - **A** is for Atomicity
  - **C** for Consistency
  - **I** for Isolation
  - **D** for Durability.



1.Atomicity

- Either entire transaction takes place at once or doesn’t happen at all.
- If any part fails, the entire transaction is rolled back to keep the database consistent.
  - ***Commit:*** If the transaction is successful, the changes are permanently applied.
  - ***Abort/Rollback:*** If the transaction fails, any changes made during the transaction are discarded.

Before: X : 500	Y : 200
Transaction T	
T1	T2
Read (X) X: = X - 100 Write (X)	Read (Y) Y: = Y + 100 Write (Y)
After: X : 400	Y : 300

- If the transaction fails after completion of T1 but before completion of T2, the database would be left in an inconsistent state.
- With Atomicity, if any part of the transaction fails, the entire process is rolled back to its original state, and no partial changes are made.

2.Consistency

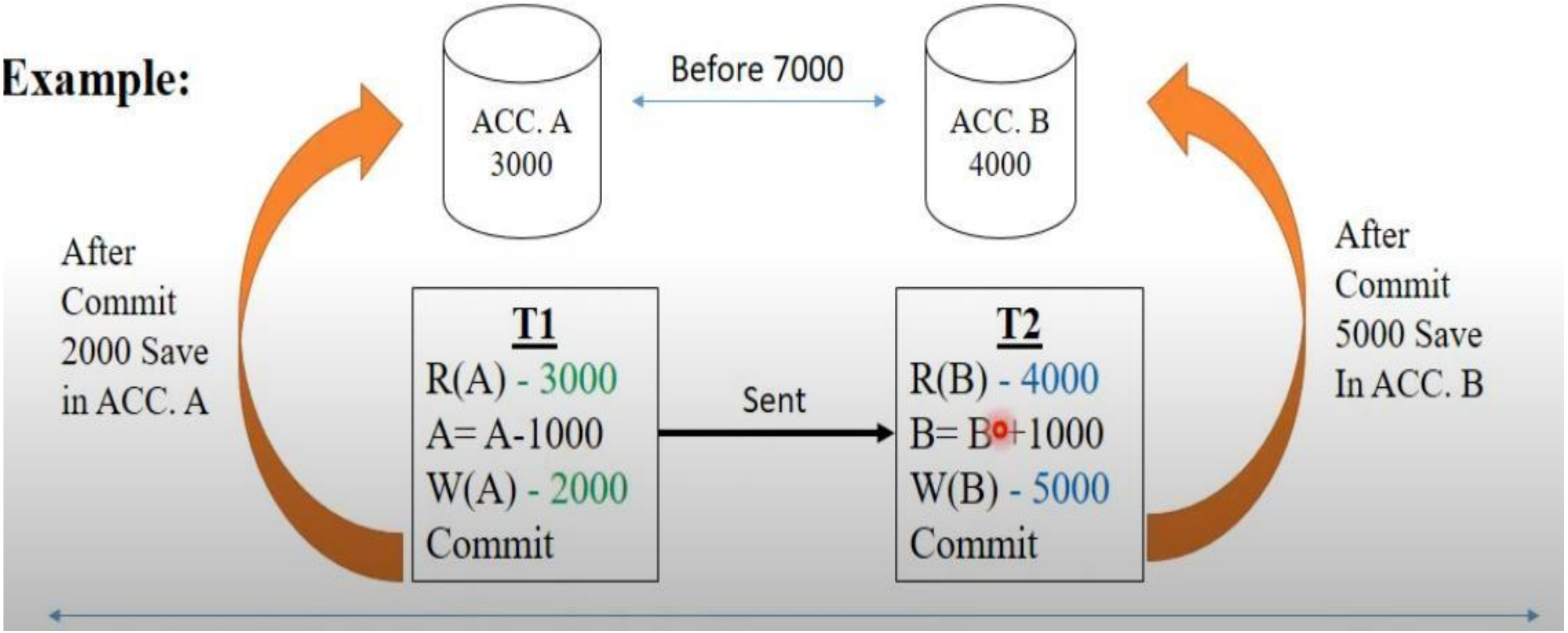
- It refers to correctness of database.
- The **Consistency property ensures that the database remains in a consistent state before and after a transaction**.
- In above example the total amount before and after transaction must be maintained correctly.

Total Amount before T occurs = 500 + 200 = 700 .

Total Amount after T occurs = 400 + 300 = 700 .

Hence Consistency property exist in this example.

Example:


ACC. A
3000
Before 7000
ACC. B
4000
After
Commit
2000 Save
in ACC. A
T1
R(A) - 3000
A= A-1000
W(A) - 2000
Commit
Sent
T2
R(B) - 4000
B= B+1000
W(B) - 5000
Commit
After
Commit
5000 Save
In ACC. B

3. Isolation

- Isolation maintains the independence of database transactions.
- The **operation on one database should start only when operation on first databse gets completed**.
- Uncommitted transactions are isolated using locking mechanisms to prevent dirty reads or lost updates.
- For example, if Transaction T1 updates a row, Transaction T2 must wait until T1 commits or rolls back, ensuring that T2 does not read unreliable data.

4. Durability

- Durability **ensures that once a transaction is committed, its changes are permanently saved, even if the system fails**.
- The data is stored in non-volatile memory, so the database can recover to its last committed state without losing data.
- Example: After successfully transferring money from Account A to Account B, the changes are stored on disk. Even if there is a crash immediately after the commit, the transfer details will still be intact when the system recovers, ensuring durability.